MILITARY SPECIFICATION SHEET

TUBING, PLASTIC, EXTRA FLEXIBLE, CONVOLUTED, ETHYLENE-TETRAFLUOROETHYLENE, CLOSE CONVOLUTION

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the tubing described herein shall consist of this document and the issue in effect of MIL-T-81914(AS).

REQUIREMENTS:

Convolution type: Helical (see 3.3)

Class:

Class 1 - Standard wall thickness
Class 2 - Thin wall thickness

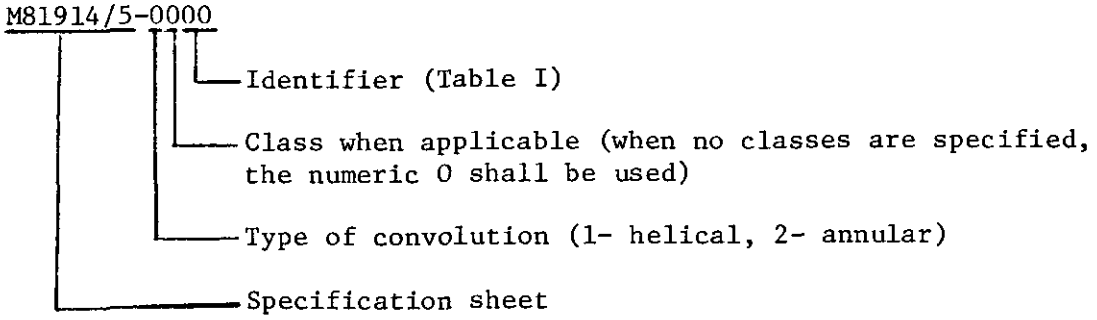
Construction details: Construction details, including available sizes, dimensions and tolerances, are located in Table I. Lengths shall be as specified by the procuring activity.

Continuous operating temperature: -67°C (-88°F) to 150°C (302°F)

Color: Unless otherwise specified, the supplied color shall be clear, natural.

Physical properties: General physical requirement values along with associated test conditions are located in Table II.

Military Part Number: Consists of the basic number of this specification sheet and a dash number as shown below:



Close convolution, Ethylene-Tetrafluoroethylene, thin wall, 0.500 inch is identified as M81914/5-1206.

* U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/6681

FSC 9330

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TABLE I

CONSTRUCTION DETAILS

IDENT-IFIER	MAXIMUM INSIDE DIAMETER	MINIMUM INSIDE DIAMETER	MAXIMUM OUTER DIAMETER	WALL THICKNESS MAXIMUM		CONVOLUTIONS PER INCH ±1		WEIGHT PER 100 FT. MAX.		MINIMUM BEND RADIUS
				CLASS 1	CLASS 2	CLASS 1	CLASS 2	CLASS 1	CLASS 2	
**01	.188	.181	.320	.018	.012	10	11	1.4	1.2	0.31
**02	.281	.273	.414	.018	.012	10	11	1.6	1.3	0.41
**03	.312	.306	.450	.018	.012	10	11	1.7	1.5	0.41
**04	.375	.359	.510	.018	.012	10	11	2.0	1.6	0.50
**05	.437	.427	.571	.018	.012	10	11	3.1	1.90	0.50
**06	.500	.480	.650	.023	.012	9	10	3.7	2.20	0.75
**07	.625	.603	.770	.023	.012	9	10	4.4	2.75	0.75
**08	.750	.725	.930	.023	.012	8	9	5.6	3.30	0.93
**09	.875	.860	1.073	.023	.012	7	8	7.1	3.85	1.25
**10	1.000	.970	1.226	.023	.012	7	8	7.6	4.40	1.25
**11	1.125	1.105	1.390	.023	.012	7	8	8.4	4.95	1.43
**12	1.250	1.205	1.539	.023	.012	6	7	9.0	5.50	1.43
**13	1.500	1.437	1.832	.023	.012	5	6	9.6	6.60	1.75
**14	1.750	1.688	2.082	.023	.012	5	6	10.7	7.70	2.00
**15	2.000	1.937	2.332	.023	.012	5	6	12.4	8.85	2.25
**16	6.000	5.870	6.120	.028		4		28.5		11.0

** The asterisks shall be replaced by convolution type and class designation

NOTE: Unless otherwise specified, all dimensions are in inches.

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TABLE II

PHYSICAL REQUIREMENTS

PROPERTY	REQUIREMENT	TEST METHOD
Construction details, inches	In accordance with Table I	4.6.1
Stress in psi at 10% strain	Class 1 - 200 to 650 Class 2 - 200 to 850	4.6.2
Specific gravity, max.	1.70	4.6.3
Crush resistance, lbs., Horizontal, min.	Class 1-12; Class 2-9	4.6.4
Low temperature flexibility	No cracking	4.6.5; 10,000 cycles
Heat shock	No dripping, flowing or cracking	4.6.6; 175 ±1°C (347 ±2°F) for 4 hours
Longitudinal change % max.	15	
Heat aging		4.6.7; 175 ±1°C (347 ±2°F) for 96 hours
Stress in psi @ 10% strain, % min. of init. value	80	
Fluid resistance		4.6.8
Stress in psi @ 10% strain, min. % of init. value	80	
Flammability	Not flammable	4.6.9
Fungus resistance	Non-nutrient 1/	4.6.10
Dielectric breakdown	10,000 volts no breakdown	4.6.11

1/ See 6.4 "Properties not tested"

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Standard Procurement: Procured as coils. Unless otherwise specified, individual lengths shall be 18 feet.

Preparing Activity
Navy - AS
(Project No. 9330-N750)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-T-81914/5(AS) Tubing, Plastic, Extraflexible, Convoluted, Ethylene-tetrafluoroethylene, Close Convolution.

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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